

SULIT
DP014
Physics 1
Semester I
Session 2024/2025
2 hours

DP014
Fizik 1
Semester I
Sesi 2024/2025
2 jam



KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **10** halaman bercetak.

*This question paper consists of **10** printed pages.*

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SULIT

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **7** questions.

Answer **all** questions.

All answers must be written in the answer booklet provided. Use a new page for each question.

The use of non-programmable scientific calculator is permitted.

ARAHAN KEPADA CALON:

*Kertas soalan ini mengandungi **7** soalan.*

*Jawab **semua** soalan.*

Semua jawapan hendaklah ditulis pada buku jawapan yang disediakan. Gunakan muka surat baharu bagi nombor soalan yang berbeza.

Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

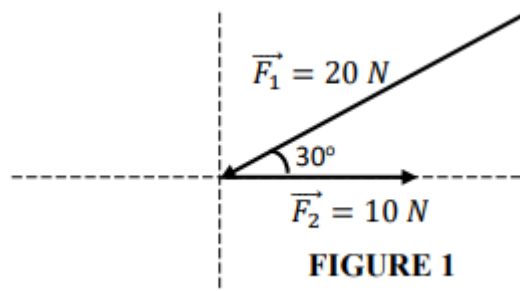
LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|--|
| 1. $v = u + at$ | 23. $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$ |
| 2. $s = ut + \frac{1}{2} at^2$ | 24. $\omega^2 = \omega_0^2 + 2\alpha\theta$ |
| 3. $v^2 = u^2 + 2as$ | 25. $\theta = \frac{1}{2}(\omega_0 + \omega)t$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 26. $\tau = rF \sin \theta$ |
| 5. $v = u - gt$ | 27. $\frac{Q}{t} = -kA \frac{\Delta T}{L}$ |
| 6. $v^2 = u^2 - 2gs$ | 28. $v_{rms} = \sqrt{\langle v^2 \rangle}$ |
| 7. $s = ut - \frac{1}{2} gt^2$ | 29. $v_{rms} = Q \frac{3kT}{m} = Q \frac{3RT}{M}$ |
| 8. $p = mv$ | 30. $\Delta U = Q - W$ |
| 9. $J = F\Delta t$ | |
| 10. $J = \Delta p = mv - mu$ | |
| 11. $f_s \leq \mu_s N$ | |
| 12. $f_k = \mu_k N$ | |
| 13. $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$ | |
| 14. $K = \frac{1}{2} mv^2$ | |
| 15. $U = mgh$ | |
| 16. $U_s = \frac{1}{2} kx^2 = \frac{1}{2} Fx$ | |
| 17. $a = \frac{v^2}{r} = r\omega^2 = v\omega$ | |
| 18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$ | |
| 19. $s = r\theta$ | |
| 20. $v = r\omega$ | |
| 21. $a_t = r\alpha$ | |
| 22. $\omega = \omega_0 + \alpha t$ | |

1. a) Calculate the momentum, p of object in SI unit if mass, $m = 2000 \text{ g}$ and velocity, $v = 300 \text{ km h}^{-1}$
- [2 marks]

- b) Two forces $F_1 = 20 \text{ N}$ and $F_2 = 10 \text{ N}$ are acting on Block A as shown in **Figure 1** below. Determine the magnitude of the resultant force acting on Block A.

[6 marks]



2. (a) A lorry initially at rest travelling along 300 m after 25 s. Calculate
- (i) final velocity of the lorry after 25 s.
 - (ii) acceleration of the lorry in 25 s.
 - (iii) If the lorry then slowing down its speed at rate 0.60 m s^{-2} , how far it is travel before it stop.

[7 marks]

- (b) A ball is thrown vertically upward with speed of 19.0 m s^{-1} from 10 m tall building. Calculate the

- (i) maximum height reached by the ball from the released point.
- (ii) time taken by the ball to reach the ground

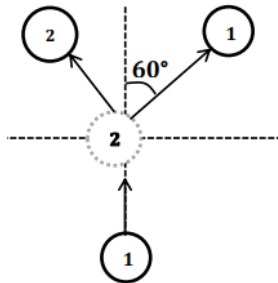
[5 marks]

3. (a) A 8 kg moves towards the west with a momentum of 30 kgms^{-1} . A 20 N force to the east acts on the body for a period of 15 s. determine the magnitude of

- (i) the impulse of the force
- (ii) the change in the momentum of a body
- (iii) the final momentum of the body
- (iv) the final velocity of the body

[5 marks]

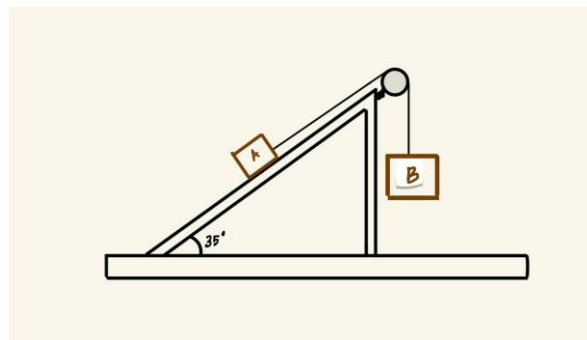
(b)

**Figure 3**

A ball 1 with mass 0.15 kg moving to the north at speed 5 ms^{-1} collide with a stationary ball 2 whose mass is 0.20 kg . After collide the ball 1 moves with speed 9 ms^{-1} . Calculate the velocity of ball 2 after collision.

[4 marks]

4. (a) **Figure 4.1** show two block A and B connected to a massless strings passing through smooth pulleys. Block A is placed on a rough inclined surface of coefficient of kinetic friction is 0.6 with a mass of 25 kg moving along rough surface. Block B moving downward experience an acceleration 0.5 ms^{-2}

**Figure 4.1**

- (i) Draw free body diagram to show all forces that act on block A.
- (ii) Calculate the frictional force acting on the block A.
- (iii) Calculate the tension of the rope that connecting block A and B.
- (iv) Calculate the mass of block B.

[9 marks]

- (b) Two boxes moving at a frictionless horizontal surface are connected by a light string as shown in **Figure 4.2**, where $m_1 = 12\text{ kg}$ and $m_2 = 25\text{ kg}$. A 60 N force is applied to the m_2 box with an angle of 45° with the horizontal surface.

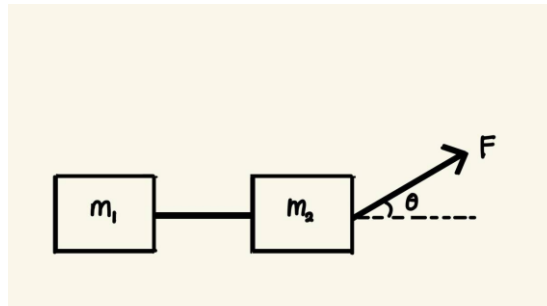


Figure 4.2

- (i) Find the acceleration of each box
- (ii) Find the tension in the string

[3 marks]

5. (a) An object of mass 450 g moves in a circle of diameter 3.0 m . It completes 5 revolutions in 2 seconds. Calculate
- (i) the angular velocity in SI unit.
 - (ii) the linear speed of the object.

[4 marks]

- (b) An object of mass 12 g is attached to the end of a string of length 50 cm . The object moves through 5 rounds per second in a circular motion on a smooth horizontal surface.
- (i) Draw a free body diagram to show all the forces acting on the object.
 - (ii) Calculate the net force acting on the object.
 - (iii) Calculate the tension in the string..

[5 marks]

6. (a) A rotating wheel moving from rest with 20 cm of radius an angular acceleration of 3.0 rad s^{-2} for 28 s . Calculate
- (i) The angular velocity
 - (ii) Tangential velocity
 - (iii) The angle through which the wheel has turned

[6 marks]

- (b) Three forces are acting on a lightweight object as shown in **FIGURE 6**. Calculate the net torque on the object at the pivoted point P.

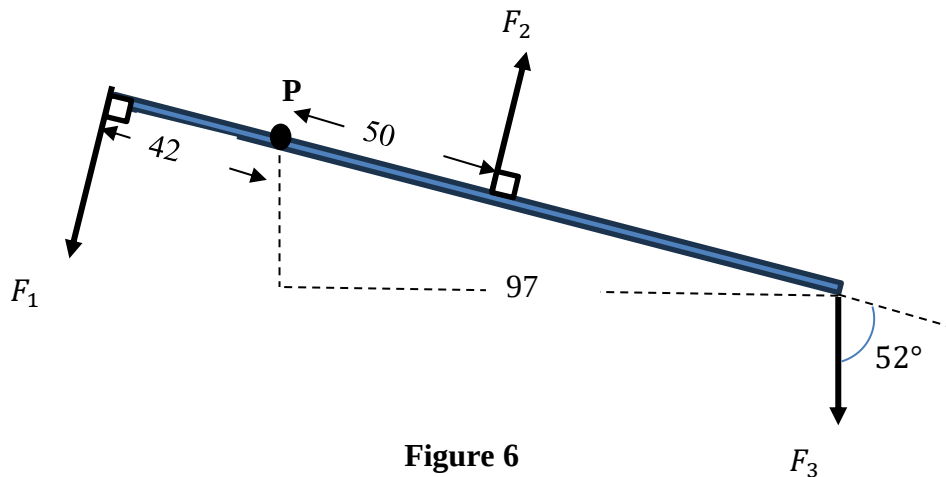


Figure 6

[6 marks]

7. (a) A copper rod has a radius of 8.0 mm and a length of 10.0 cm. One end of the rod is immersed in boiling water, while the other end is immersed in ice. Calculate rate of heat flow through the rod if Thermal Conductivity of the Copper is $385.0 \text{ W m}^{-1} \text{ K}^{-1}$.

[5 marks]

- (b) (i) The rms speed of the atoms of a monatomic gas at a temperature of 20°C is 645 ms^{-1} . Find the mass of an atom of the gas.
- (ii) The rms speed of hydrogen molecules at a particular temperature is 1320 ms^{-1} . What is the rms speed of nitrogen molecules at the same temperature? (The molar mass of hydrogen is 2.0 g mol^{-1} and for nitrogen is 28 g mol^{-1})

[5 marks]

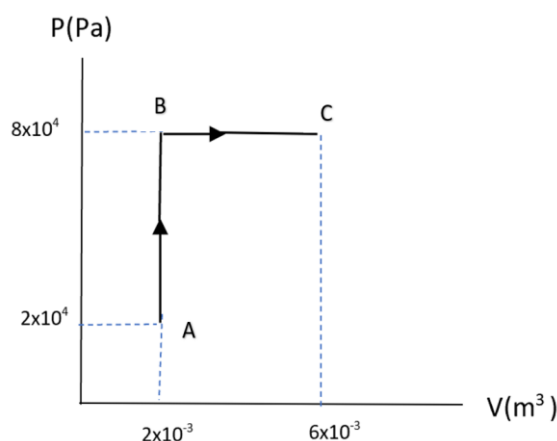


FIGURE 7

- (c) (i) **FIGURE 7** shows a P-V graph for a series of thermodynamics processes, ABC.

In process AB and BC, 180 J and 250 J heat are added to the system respectively. Work done by the system in BC process is 120 J. Calculate the changes in internal energy during process ABC.

[5 marks]

- (ii) As an ideal gas is compressed isothermally, 225 J of heat released. Calculate work done and state either work is done by the system or on the system for this thermodynamic process.

[3 marks]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT